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United States Patent Application Publication

20250277944

Kind Code

A1

Publication Date

September 04, 2025

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Dust Plug for Adapter Pre-populated with Fiber Optic Connector Housing

Abstract

A dust plug for use in a pre-populated adapters has a main body having a top wall, a bottom wall, and two opposing side walls extending between the top wall and bottom wall. The dust plug also includes at least one pair of latches in a front portion of the main body and each of the at least one pair of latches may have an increased thickness at a forward end thereof. The area of increased thickness is sized to engage a receptacle in an inner surface of the housing when the dust plug is attached to the housing. Other versions contemplate not having the increased thickness. The dust plug preferably includes a grip at the back end.

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Family ID: 78082852

Appl. No.: 19/210093

Filed: May 16, 2025

Related U.S. Application Data

parent US continuation 18390746 20231220 parent-grant-document US 12326597 child US 19210093

parent US continuation 17231829 20210415 parent-grant-document US 11860426 child US 18390746

parent US continuation-in-part 17227750 20210412 parent-grant-document US 11474307 child US 17231829

us-provisional-application US 63075335 20200908

us-provisional-application US 63010216 20200415

Publication Classification

Int. Cl.: G02B6/38 (20060101)

U.S. Cl.:

CPC G02B6/3893 (20130101); G02B6/3825 (20130101); G02B6/3831 (20130101);
G02B6/3849 (20130101); G02B6/3897 (20130101);

Background/Summary

REFERENCE TO RELATED CASE [0001] This application claims priority under 35 U.S.C. § 119 (e) to U.S. provisional application No. 63/075,335 filed on Sep. 8, 2020, and to U.S. Provisional Application No. 63/010,216, titled “AN INTERCONNECT SYSTEM AND METHODS OF INSTALLING THE SAME,” filed Apr. 15, 2020, and under 35 U.S.C. § 120 to U.S. patent application Ser. No. 17/231,829 filed on Apr. 15, 2021; to U.S. patent application Ser. No. 17/227,750 filed on Apr. 12, 2021; and to U.S. patent application Ser. No. 18/390,746, filed on Dec. 20, 2023, the contents of are hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

[0002] Interconnect systems in high-density environments, such as in hyperscale data centers, utilize a fiber optic cable being pre-terminated with some of the components of optical connectors, and at least one adapter panel having adapters pre-populated (i.e., pre-installed) with other components of those optical connectors. The components terminating the fiber optic cable and the components pre-installed on the adapters can be combined during installation of the interconnect system, resulting in functionable optical connectors populating the adapter panel(s). The combining can occur without removing the components that are pre-installed on the adapters. For multi-fiber connections using, e.g., an MT-ferrule, these pre-installed components include a housing and an outer sleeve. The pre-installed housing can hold the ferrule with the terminated fibers therein, and the outer sleeve can slide over the housing so that adapter latches can attach to the housing. The adapter itself is attached to an adapter panel having a multitude of such adapters. The MT-ferrule can be pushed into the housing from one side of the adapter (field-termination side), e.g., using a ferrule push, to complete the connection. Typically, on the opposite side of the adapter, another MPO connector may be present. This setup is described in the U.S. Provisional Application No. 63/010,216, titled “AN INTERCONNECT SYSTEM AND METHODS OF INSTALLING THE SAME,” filed Apr. 15, 2020 (“the '216 application”).

[0003] When the adapter panel pre-populated with the housing and outer sleeve is not in use, the housing and the adapter on the field termination side can be exposed to dust and debris, contaminating the components. To prevent the dust and debris from entering the system, a dust plug is provided to cover an opening of the housing when the fiber optic ferrule is not inserted therein. On a side opposite the housing and the sleeve, i.e., the side on the other side of the panel, a regular MPO connector may be removably attached. When the MPO connector on that side is not present, the adapter may be closed off using an MPO side adapter dust plug, which is standard in the industry. A dust plug for the housing and outer sleeve is provided.

SUMMARY OF THE INVENTION

[0004] According to one aspect, the present invention is directed to a dust plug for a pre-populated adapter having a housing and a sleeve, that includes a main body having a top wall, a bottom wall, and two opposing side walls extending between the top wall and bottom wall, and at least one pair of latches in a front portion of the main body and each of the at least one pair of latches having an

increased thickness at a forward end thereof, wherein the increased thickness is sized to engage a receptacle in an inner surface of the housing when the dust plug is attached to the housing.

[0005] In some embodiments, the pair of latches further comprises a pair of vertical latches and a pair of horizontal latches, the pair of vertical latches oriented perpendicular to the pair of horizontal latches at the front portion of the main body.

[0006] In some embodiments, there is a shoulder at least partially surrounding the main body to form a stop surface configured to engage the housing and prevent further forward movement of the dust plug relative to the housing.

[0007] In some embodiments, there is a grip on a rearmost portion of the dust plug.

[0008] In some embodiments, the increased thickness comprises a plurality of bumps.

[0009] In yet another aspect, there is an interconnect system having at least one adapter panel for managing interconnections, each adapter panel including a plurality of adapters, the system including a plurality of connector housings, each of the plurality of connector housings attached to and pre-installed onto a respective adapter of the plurality of adapters, a plurality of dust plugs, each of the plurality of dust plugs attachable to a respective one of the plurality of connector housings, each of the plurality of dust plugs having a main body with a top wall, a bottom wall, and two opposing side walls disposed between the top wall and bottom wall, and at least one pair of latches in a front portion of the main body and having an increased thickness at a forward end thereof, the increased thickness is sized to engage a receptacle in an inner surface of a respective housing when the dust plug is inserted into the housing.

[0010] In some embodiments, the at least one pair of latches comprises two pair of latches, a pair of vertical latches and a pair of horizontal latches, the pair of vertical latches oriented perpendicular to the pair of horizontal latches at the front portion of the main body.

[0011] In some embodiments, there is a shoulder at least partially surrounding the main body to form a stop surface configured to engage the housing and prevent further forward movement of the dust plug relative to the housing.

[0012] In yet another aspect, there is a dust plug for a pre-populated adapter having a housing and a sleeve that includes a main body having a top wall, a bottom wall, and two opposing side walls extending between the top wall and bottom wall, and at least one pair of latches in a front portion of the main body, wherein the at least one pair of latches engages an inner surface of the housing when the dust plug is attached to the housing.

[0013] It is to be understood that both the foregoing general description and the following detailed description of the present embodiments of the invention are intended to provide an overview or framework for understanding the nature and character of the invention as it is claimed. The accompanying drawings are included to provide a further understanding of the invention, and are incorporated into and constitute a part of this specification. The drawings illustrate various embodiments of the invention and, together with the description, serve to explain the principles and operations of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a side perspective view of one embodiment of a dust plug inserted into a housing installed into an adapter according to the present invention;

[0015] FIG. 2 is a side perspective view the dust plug and a fiber optic ferrule that can be inserted into the housing in FIG. 1;

[0016] FIG. 3 is a perspective view from the upper right of the dust plug of FIG. 2;

[0017] FIG. 4 is an end view of the dust plug in FIG. 2;

[0018] FIG. 5 is a cross sectional view of the dust plug in FIG. 2;

[0019] FIG. **6** is a cross sectional view of the housing in FIG. **2** showing the internal groove for the dust plug;

[0020] FIG. **7** is a cross sectional view of the adapter, outer sleeve, housing, dust plug according to the present invention, and the standard MPO side adapter dust plug in FIG. **1**;

[0021] FIG. **8** is a cross section view of the dust plug, housing and outer sleeve in FIG. **2**;

[0022] FIG. **9** is a cross section view of the dust plug, housing and outer sleeve in FIG. **2** at a 90 degree angle from FIG. **8**; and

[0023] FIG. **10** is a partial cross section view of the dust plug, housing and outer sleeve in FIG. **1**.

DETAILED DESCRIPTION OF THE INVENTION

[0024] Reference will now be made in detail to the present preferred embodiment(s) of the invention, examples of which are illustrated in the accompanying drawings. Whenever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts.

[0025] Illustrated in FIGS. **1** and **2** is one embodiment of a dust plug **100** according to the present invention. The dust plug **100** is illustrated in FIG. **1** as being inserted into a housing **102**, along with a sleeve **104**, all of which are inserted one side of an adapter **106**. On the other side of the adapter **106** is an MPO side adapter dust plug **108**. As noted above, the dust plug **108** is known. Alternatively, there may also be a regular MPO connector may be removably attached to the adapter **106**. The adapter **106** is installed in a panel that receives a number of these adapters **106**. It should also be noted that the housing **102** and the sleeve **104** are not meant to be removed from the adapter **106**, but may be if so desired. The housing **102** and the sleeve **104** are able to receive either a fiber optic ferrule **110** and a ferrule push **112** or a dust plug **100**. See FIG. **2**. The ferrule push is described in Applicant's co-pending application Ser. No. 17/251,076, filed on Dec. 20, 2020 and entitled "Ferrule Push." The contents of this co-pending application are incorporated herein by reference. The presence of the either the fiber optic ferrule **110** and a ferrule push **112** or the dust plug **100** prevents dust, debris and other contaminants from entering into the adapter **106** through the housing **102** and the sleeve **104**.

[0026] The dust plug **100** will now be discussed with reference to FIGS. **3-5** and **6-9**. The dust plug **100** has a main body **120**, a top wall **122**, a bottom wall **124**, and two opposing side walls **126,128** extending between the top wall **122** and bottom wall **124**. The dust plug **100** also has at least one pair of latches **130** in a front portion **132** of the main body **120**.

[0027] Applicant notes that the term "front" or "forward" means that direction where the fiber optic ferrule would meet with another fiber optic ferrule or device, while the term "rear" or "rearward" is used to mean the direction from which the optical fibers enter into the fiber-optic ferrule or fiber optic connector. In the present application, the dust plug **100** has will therefore have a front and a rear, the front will be inserted into the housing and sleeve first. Thus, in FIG. **3**, the "front" of the dust plug (and also a fiber optic ferrule or connector) is on the left side of the figure and pointing out of the figure. The "rear" or "back" is that part of the dust plug (or fiber optic connector) is on the right side of the page and "rearward" and "backward" is toward the right and into the page.

[0028] The latches **130** could be bumps or protrusions as illustrated best in FIGS. **3** and **5**. The latches **130** need not cover the entirety of the front portion **132** of the main body **120**. However, there may also be a thickening of the front portion **132** to engage a receptacle **134** on an inner surface **136** of the housing **102** as detailed more below. The latches **130** may also be more than one bump or protrusion, i.e., two or three bumps.

[0029] The dust plug **100** has two cut-outs or gaps **138** in the top wall **122** and the bottom wall **124** at the front portion **132**, thereby separating the top wall **122** and the bottom wall **124** from the two opposing side walls **126,128**. This means that the top wall **122** and the bottom wall **124** are cantilevered. It also means that the two opposing side walls **126,128** are also cantilevered, allowing all of the walls to flex when a force is applied. Typically, the force would be applied from the outside of the dust plug **100**.

[0030] It should be noted that there could be cut-outs or gaps in the two opposing side walls

126,128 rather than in the top wall **122** and the bottom wall **124** as illustrated in the figures. This would also provide the cantilevered effect for the front portion **132** of the main body **120**. The latches **130** on the side walls **126,128** are preferably larger and thicker than those on the top wall **122** and the bottom wall **124**, but they could all be of the same size. Additionally, the latches **130** on the side walls **126,128** could also be simply an increased thickness at the front portion **132**. [0031] The main body **120** preferably has a shoulder **140** that extends around the circumference of the main body **120**. However, the shoulder **140** could only extend around a portion of the main body, have multiple sections, etc. The shoulder **140** provides a stop for inserting the dust plug **100** into the housing **102**. See FIG. 2. The shoulder **140** has a forward facing surface **142** that engages the rearward facing surface **102a** of the housing **102**. See FIG. 1. This prevents the dust plug **100** from being inserted too far relative to the housing **102**. Preferably the latches **130** are positioned on the main body **120** so that they engage a receptacle or inner surface of the housing **102** as described below with reference to FIGS. 7-9.

[0032] At the rear **144** of the main body **120** is a grip **146**. The grip **146** is preferably used to pull the dust plug **100** from the housing **102** and to insert the dust plug into the housing **102**. The grip **146** has a rectangular shape, but could have any configuration to allow a user to grasp the dust plug **100**. While there may be other structures behind the grip **146**, is preferred that the grip **146** be the rearmost structure on the dust plug **100**.

[0033] The top wall **122** and the bottom wall **124** and the two opposing side walls **126,128** form an opening **150** in the main body **120**. The opening **150** provides an area for the top wall **122**, the bottom wall **124**, and the two opposing side walls **126,128** to flex into when a force is applied to the outside of the dust plug **100**. The opening **150** may also include a spine **160**. The spine **160** provides for strength, support, and rigidity to the top wall **122** and the bottom wall **124**. The spine **160** preferably extends from a back wall **162** of the opening towards the front portion **132**. The spine **160** as illustrated in FIGS. 3, 5 and 7-8 show the spine **160** extending to the very front of the main body **120**. However, the spine **160** could end anywhere along the length of the opening **150**. Additionally, the spine **160** has an opening **164** that has a v-shape, with the vertex oriented toward the back wall **162**. The shape of the opening **164** could take any applicable form, e.g., rectangular, oval, circular, etc. The spine **160** is preferably arranged in the middle of the opening between the two side walls, **126,128**, but could be located closer to one of the side walls if so desired. There may also be other openings in the spine, for example, at the back wall **162**, or between the back wall **162** and the front portion **132**. The thickness of the spine **160** may also vary from that shown, depending on the length and positioning of the spine **160** in the opening **150**.

[0034] Turning to FIG. 6, there is a cross section of the housing **102**. The housing **102** has a top wall **170** and a bottom wall **172**, which are joined by side walls **174** (the other side wall is not shown in this cross section view, although it has the same construction). The side wall **174** has a receptacle **176** in the side wall **174** (and the other side wall would also have the same receptacle-see also FIGS. 7 and 8. The receptacle **176** receives the latch **130** on the side wall **126**. While the receptacle **176** is illustrated as an opening through the side wall **172**, it may only be a depression.

[0035] The walls **170,172,174** of the housing **102** form an opening **178**, into which the dust plug **100** or the fiber optic ferrule **110** and a ferrule push **112** can be inserted. In an inner surface **180** of the top wall **170** (and in communication with the opening **178**) of the housing **102** is a groove or recessed portion **182**. This groove **182** may function as a key way for the fiber optic ferrule **110** and a ferrule push **112**. It is also may receive the latch **130** from the top wall **122** of the dust plug **100**.

[0036] With reference to FIGS. 7-10, the combination of the dust plug **100** with the adapter **106** and the housing **102** with the sleeve **104** will be discussed. On the right side of FIG. 7 is the an MPO side adapter dust plug **108** and it has been inserted into the adapter **106**. In the typical installation of these adapters, there are a number of the adapters **106** that are inserted into a number of panels to form the interconnect system. On the left side of the adapter **106** is a sleeve **104** and a housing **102**. The adapter **106** engages a portion of the housing **102** to retain the sleeve **104** and the

housing **102** therein. The sleeve **104** is slidably engaging the housing **102**, with a spring **190** disposed between the two. The dust plug **100** is then inserted into the housing **102**. As the dust plug **100** is inserted into the housing **102**, the latches **130** are flexed inward and engage the inner surface of the opening **150**. Once the dust plug **100** is fully inserted, the shoulder **140** engages the rearward facing surface **102a** of the housing **102**. The latches **130** on the side walls **126,128** engage the receptacles **176** in the housing **102**. This engagement reduces the chance of the dust plug **100** from falling out of the housing **102**. The latches **130** on the top wall **122** and the bottom wall **124** frictionally engage the inner surface of the top wall **170** and the bottom wall **172**. See FIG. **8** showing only the dust plug **100** with the housing **102** and the sleeve **104**.

[0037] There may or may not be receptacles similar to receptacles **176** in the top wall **170** and the bottom wall **172**. However, as noted above, there may be a groove **182** in the housing **102**. See FIG. **9** showing the housing **102** with the groove **182** therein. In that case, the side walls **126,128** and the bottom wall **124** would engage the inside of the housing **102**, with the latch **130** on the bottom wall **124** frictionally engaging the housing.

[0038] It will be apparent to those skilled in the art that various modifications and variations can be made to the present invention without departing from the spirit and scope of the invention. Thus it is intended that the present invention cover the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents.

Claims

1. An adapter panel for managing interconnections, each adapter panel including a plurality of adapters, each adapter in the plurality of adapters comprising: a first side with a first opening to receive a fiber optic connector or a dust plug when the fiber optic connector is not present on the first side; and a second side with a connector housing attached thereto and pre-installed onto a respective adapter of the plurality of adapters, the connector housing having an inner surface, wherein the first side and the second side are on opposite sides of the adapter panel; wherein on the second side, a dust plug is attached to the connector housing, the dust plug having a main body with a top wall, a bottom wall, two opposing side walls disposed between the top wall and bottom wall, and at least one pair of latches in a front portion of the main body to engage the inner surface of the connector housing, and wherein the connector housing includes a slidable sleeve non-removable therefrom but movable longitudinally.
2. The adapter panel according to claim 1, wherein the at least one pair of latches comprises two pairs of latches including a pair of vertical latches and a pair of horizontal latches, the pair of vertical latches oriented perpendicular to the pair of horizontal latches at the front portion of the main body.
3. The adapter panel according to claim 1, further comprising: a multi-fiber ferrule received within the connector housing only when the dust plug is not attached to the connector housing, the connector housing interchangeably receiving the multi-fiber ferrule or the dust plug one at a time.
4. The adapter panel according to claim 1, wherein the at least one pair of latches has an increased thickness at a forward end thereof to engage a receptacle or a depression in the inner surface of the connector housing when the dust plug is inserted thereinto.
5. The adapter panel according to claim 1, wherein the connector housing pre-installed onto the respective adapter includes the slidable sleeve at least partially surrounding the connector housing.
6. The adapter panel according to claim 1, wherein the inner surface of the connector housing has a receptacle or a depression in a side wall of the connector housing to receive the at least one pair of latches of the dust plug.
7. The adapter panel according to claim 1, wherein the connector housing is not removable from the second side.
8. The adapter panel according to claim 1, wherein the connector housing is an MPO housing.

9. The adapter panel according to claim 1, wherein the dust plug does not contact the slidable sleeve.
10. The adapter panel according to claim 1, wherein the dust plug further comprises a shoulder at least partially surrounding the main body to form a stop surface configured to engage the connector housing and prevent further forward movement of the dust plug relative to the connector housing.
11. The adapter panel according to claim 1, wherein the dust plug further comprises a grip on a rearmost portion of the dust plug.
12. The adapter panel according to claim 1, wherein the dust plug further comprises a spine disposed centrally within an opening in the main body of the dust plug and extending from the top wall to the bottom wall.
13. The adapter panel according to claim 12, wherein the spine has a cut-out at a forward portion thereof.
14. The adapter panel according to claim 1, wherein there is a gap between a portion of the top wall and each of the side walls and a gap between a portion of the bottom wall and the side walls of the dust plug.
15. An adapter panel for managing interconnections, each adapter panel including a plurality of adapters, each adapter in the plurality of adapters comprising: a first side to receive a multi-fiber fiber optic connector or a dust plug; and a second side with a connector housing attached thereto and non-removably pre-installed onto said each adapter of the plurality of adapters, the connector housing having an inner surface, wherein the first side and the second side are on opposite sides of the adapter panel; wherein the second side includes a dust plug attached to the connector housing and having a main body with a top wall, a bottom wall, two opposing side walls disposed between the top wall and bottom wall, and at least one pair of latches in a front portion of the main body and having an increased thickness to engage a receptacle or a depression of the inner surface of the connector housing, and wherein the connector housing includes a slidable sleeve non-removable therefrom but movable longitudinally thereupon.
16. The adapter panel according to claim 15, wherein the at least one pair of latches comprises two pairs of latches including a pair of vertical latches and a pair of horizontal latches, the pair of vertical latches oriented perpendicular to the pair of horizontal latches at the front portion of the main body.
17. The adapter panel according to claim 15, further comprising: a multi-fiber ferrule received within the connector housing when the dust plug is not inserted into the connector housing, the connector housing interchangeably receiving the multi-fiber ferrule or the dust plug at a time.
18. The adapter panel according to claim 15, wherein the dust plug further comprises a grip on a rearmost portion of the dust plug.
19. The adapter panel according to claim 15, wherein the connector housing has a slidable sleeve that remains attached to the adapter but is movable longitudinally.
20. An interconnect system having the adapter panel of claim 15.
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